## 1. Data Preparation, Exploration & Analysis

### A. Data Cleansing & Feature Engineering

Duplicate Handling

- Verified that each walk-in visit is uniquely identified using `Visit ID`.
- Remove duplicate data

Data Type Standardization

- Converted waiting time fields (Time to Triage, Time to Medical Professional, Total Wait Time) to numeric (minutes) to enable aggregation and statistical analysis.
- Standardized Visit Date fields to Date/Time format to support temporal analysis by hour, arrival period, and day of week.

Outlier Identification (Non-destructive)

- Created an outlier flag (`TotalWait_OutlierFlag`) to classify extreme waiting times (>300 minutes) without deleting records.

Feature Engineering

- Derived Arrival Time categories (Early Morning, Late Morning, Afternoon, Evening, Night) to support time-based analysis.
- Created Shift classification (Day / Night) to evaluate operational differences.

### B. Data Structuring

To support multi-dimensional ER analysis:

- Retained the dataset in a fact-style structure, where each row represents one walk-in ER visit.
- Time-based categorical columns (Arrival Time, Shift, Day of Week) were engineered to enable clean slicing without requiring additional dimension tables.
- A dedicated Measures table was created to store all DAX measures separately, improving model readability and presentation quality.

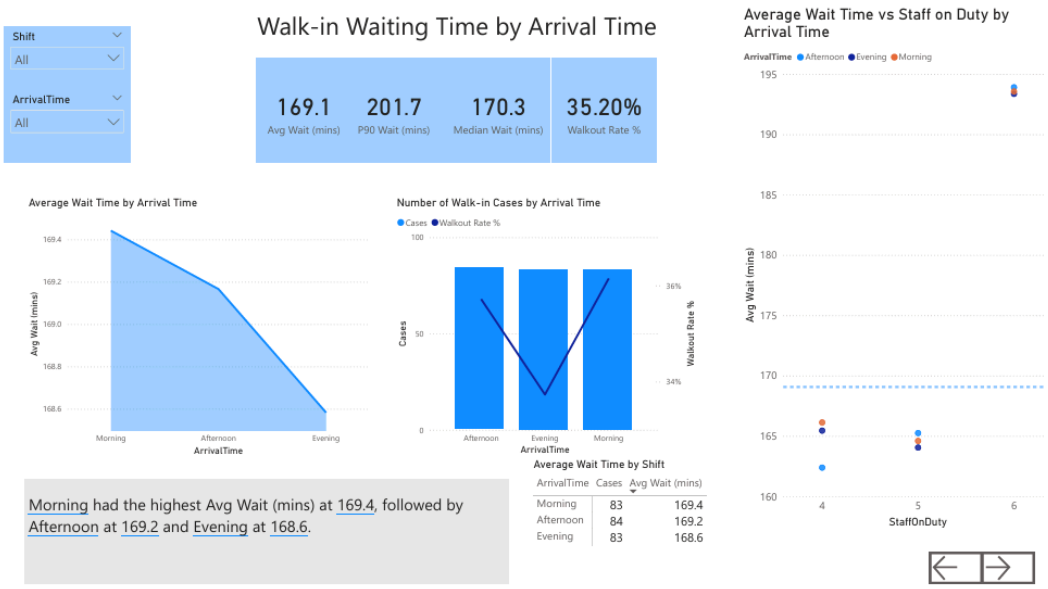
### C. Data Modeling

- Implemented a single-fact data model focused on ER walk-in visits to avoid unnecessary join complexity.
- All KPIs and analytical measures were created using DAX and evaluated dynamically through slicers (Arrival Time, Shift, Triage Level).
- Relationships were intentionally kept minimal to reduce ambiguity and ensure interpretability.

## 2. Data Exploration & Visual Analysis

Purpose: To evaluate how long ER walk-in patients wait before receiving initial clinical assessment and treatment, and to identify patterns, bottlenecks, and service impact.

### Page 1: Walk-in Waiting Time by Arrival Time




### 1. KPI Cards (Top Row)

**Visuals:** Average Wait (mins), Median Wait (mins), 90th Percentile Wait (mins), Walkout Rate (%)
**Purpose:**These KPIs provide a high-level summary of ER walk-in waiting performance. Average and median waiting times describe typical patient experience, while the 90th percentile highlights worst-case delays. The walkout rate quantifies the impact of prolonged waiting on patient behaviour.

### 2. Line Chart – Average Wait Time by Arrival Time

**Purpose:** This visual shows how average waiting time varies depending on when patients arrive (Morning, Afternoon, Evening).

**Why it matters:** This highlights time-based congestion and helps identify periods where operational pressure may be highest.

### 3. Clustered Column + Line Chart – Number of Walk-in Cases and Walkout Rate by Arrival Time

**Purpose:**This visual compares patient volume (number of cases) with walkout rate across arrival times.

**Why it matters:**This indicates that long waiting times are **not driven purely by demand**, suggesting inefficiencies in capacity or workflow.

### 4. Bar Chart – Average Wait Time by Shift

**Purpose:**This visual compares average waiting time between day and night shifts.

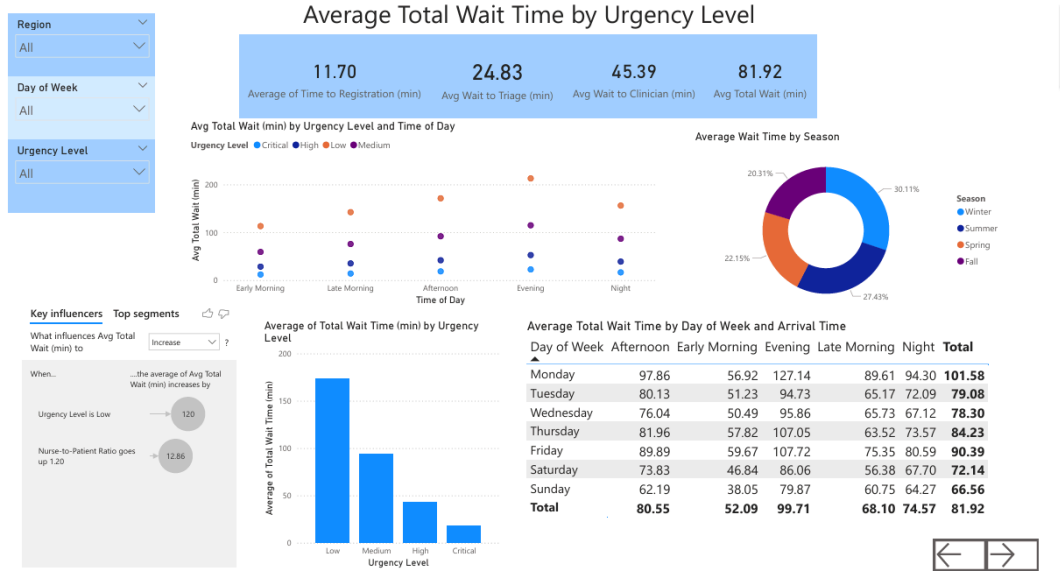
**Why it matters:** It supports staffing-related hypotheses and informs resource allocation decisions.

### 5. Scatter Plot – Average Wait Time vs Staff on Duty (by Arrival Time)

**Purpose:**This diagnostic visual explores the relationship between staffing levels and waiting time.

**Why it matters:** This suggests staffing adequacy as a potential bottleneck contributing to prolonged waits.

## Page 2: Waiting Time by Urgency and Care Stage




### 6. KPI Cards – Care Process Timing

**Visuals:**Average Time to Registration, Average Time to Triage, Average Time to Clinician, Average Total Wait Time
**Purpose:**These KPIs decompose the total waiting time into key stages of the ER care process.

**Why it matters:**They distinguish between **assessment delays** (triage) and **treatment delays** (clinician), directly answering the research question.

**7. Bar Chart – Average Total Wait Time by Urgency Level**

**Purpose:**This visual shows how waiting time differs across urgency levels (Low, Medium, High, Critical).

**Why it matters:**This confirms that the triage system prioritises critical cases appropriately, while also highlighting long waits for lower-acuity patients.

**8. Scatter / Line Chart – Avg Total Wait Time by Urgency Level and Time of Day**

**Purpose:**This visual examines how urgency and time of day interact to affect waiting time.

**Why it matters:**It highlights compounding effects where operational pressure and patient urgency intersect.

**9. Matrix – Average Total Wait Time by Day of Week and Arrival Time**

**Purpose:**This matrix provides a detailed breakdown of waiting time patterns across days and arrival periods.

**Why it matters:**This helps identify recurring bottlenecks rather than isolated incidents.

**10. Key Influencers (AI Visual) – Drivers of Waiting Time**

**Purpose:**This AI-assisted visual identifies variables most strongly associated with increased waiting time.

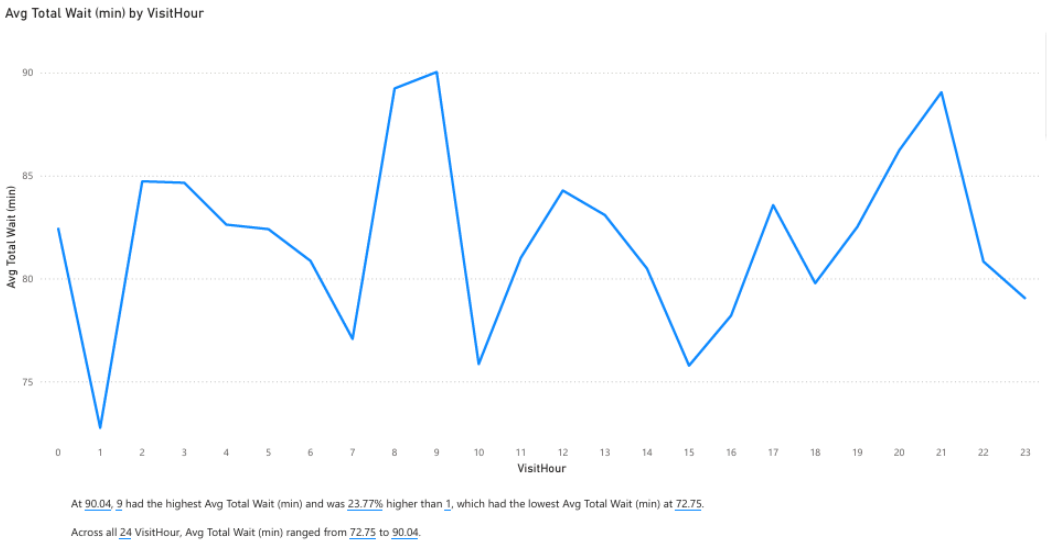
**Why it matters:**
It supports hypothesis generation and demonstrates advanced analytical capability using AI features.

**11. Donut Chart – Average Wait Time by Season**

**Purpose:**This visual shows seasonal variation in waiting time.

**Why it matters:** It provides contextual understanding but is secondary to operational drivers.

**Page 3: Visit Hour Analysis**




12. Line Chart – Average Total Wait Time by Visit Hour

Purpose:This visual shows how waiting time changes across the 24-hour clock.

What it reveals:Waiting time peaks during specific hours, indicating intraday congestion.

Why it matters: It supports fine-grained staffing and scheduling decisions.

### 3. Key Insights & Bottleneck Identification

- Waiting times are consistently high across all arrival periods, with morning walk-ins facing the longest delays.
- Patient volume alone does not explain waiting time variability, pointing to operational inefficiencies.
- Staffing levels show a negative relationship with waiting time, identifying workforce allocation as a bottleneck.

### 4. Technical Documentation (DAX Measures)

- **Case Volume :** Cases = COUNT(er_data[Visit ID]) Counts total walk-in visits for workload analysis.
- **Average Waiting Time =** Avg Wait (mins) = AVERAGE(er_data[Total Wait Time (min)]) Primary performance KPI representing overall ER waiting experience.
- **Median Waiting Time=** Median Wait (mins) = MEDIAN(er_data[Total Wait Time (min)]) Represents the typical patient experience, robust to extreme outliers.
- **90th Percentile Waiting Time =** P90 Wait (mins) =PERCENTILEX.INC( er_data, er_data[Total Wait Time (min)], 0.9) Captures worst-case but realistic waiting scenarios.
- **Walkout Rate =** Walkout Rate % = DIVIDE([Walkouts], [Cases]) Measures service impact caused by excessive waiting.